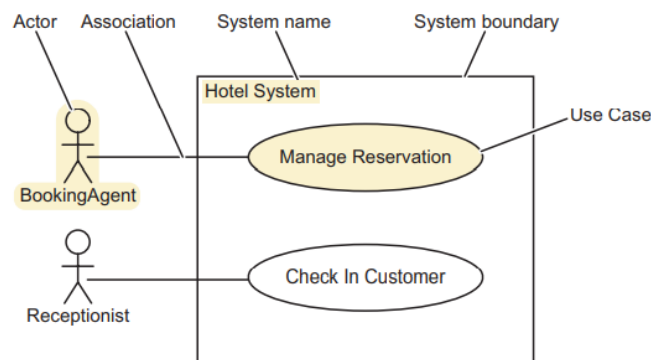


REVIEW: USE CASE DIAGRAM

Identifying the Elements of a Use Case Diagram

A Use Case diagram shows the relationships between actors (roles) and the goals they wish to achieve.

A physical job title can assume multiple actors (roles).



Following are reasons a Use Case diagram is necessary:

- A Use Case diagram enables you to identify—by modeling—the high-level **functional requirements** (FRs) that are required to satisfy each user's goals.
- The client-side stakeholders need a big picture view of the system.
- The use cases form the basis from which the detailed FRs are developed.
- Use cases can be prioritized and developed in order of priority.
- Use cases often have minimal dependencies, which enables a degree of independent development.

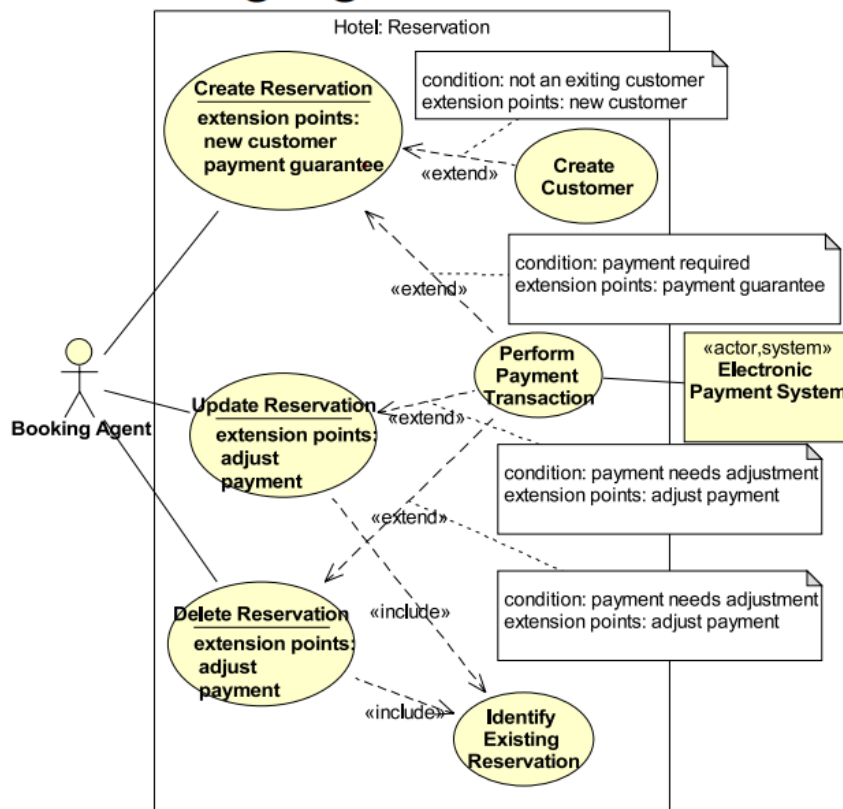
Non-Functional Requirements (NFRs)

- Non-functional requirements (NFRs) define the qualitative characteristics of the system. As in an animal, the NFRs describe strength, speed, and agility of the internal features of the animal. How fast can the animal move? How much weight can the animal carry?
- Any adverbial phrase can be an NFR.

NFRs: Examples

- NFR1: The system must support 200 simultaneous users in the Web application.
- NFR2: The process for completing any reservation activity must take the average user no more than 10 minutes to finish.
- NFR3: The capacity of reservation records could grow to 2,600 per month.
- NFR4: The Web access should use the HTTPS transport layer when critical customer information is being communicated.
- NFR5: The numerical accuracy of all financial calculations (for example, reports and customer receipts) should follow a 2-significant-digit precision with standard rounding of intermediate results.

Packaging the Use Case Views



Use Case Scenario

Form Element	Description
Use Case Name	Create Reservation
Description	The Customer requests a reservation for hotel rooms for a date range. If all the requested rooms are available, the price is calculated and offered to the Customer. If details of the customer and a payment guarantee are provided, the reservation will be confirmed to the Customer.
Actor(s)	Primary: Booking Agent, Online Booker, Travel Agent System Secondary: None Note: Primary actors are proxies for the Customer.
Priority	Must have: Essential to this system

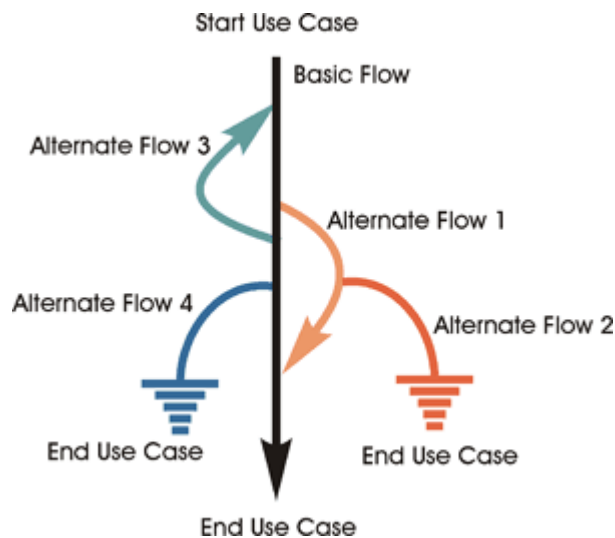
Risk	High: Primarily because of the complexity of identifying if rooms are available and the number of different actor roles that can use this use case.
Trigger	A Customer wishes to make a reservation in the hotel.
Pre-conditions	At least one room exists in the hotel. Primary Actor can be identified.
Post-conditions	One reservation is added. Payment guarantee details are recorded.
Non-Functional Requirements	<i>NFR1 (Simultaneous Users)</i> <i>NFR2 (Duration of Use Case)</i> <i>NFR4 (Web Security)</i> <i>NFR6 (System Availability)</i> <i>NFR8 (Max Time for Room Availability Search)</i>
Notes	A fast method of checking room availability is still under investigation.

Flow of Events	1: Use case starts when Customer requests to create a reservation 2: Customer enters types of rooms, arrival date, and departure date [A1] [A2] 2.1: System creates a reservation and reserves rooms applying BR3 [A3] 2.3: System calculates quoted price applying BR4 2.3.1 System records quoted price 2.4: System notifies Customer of reservation details (including rooms and price) 3: Customer accepts rooms offered [A5] 3.1: Extension Point (new customer) [A6] 3.2: Extension Point (payment guarantee) [A7] 3.3: System changes reservation status to “confirmed” 3.4: System notifies Customer of confirmed reservation details 4: Use case ends
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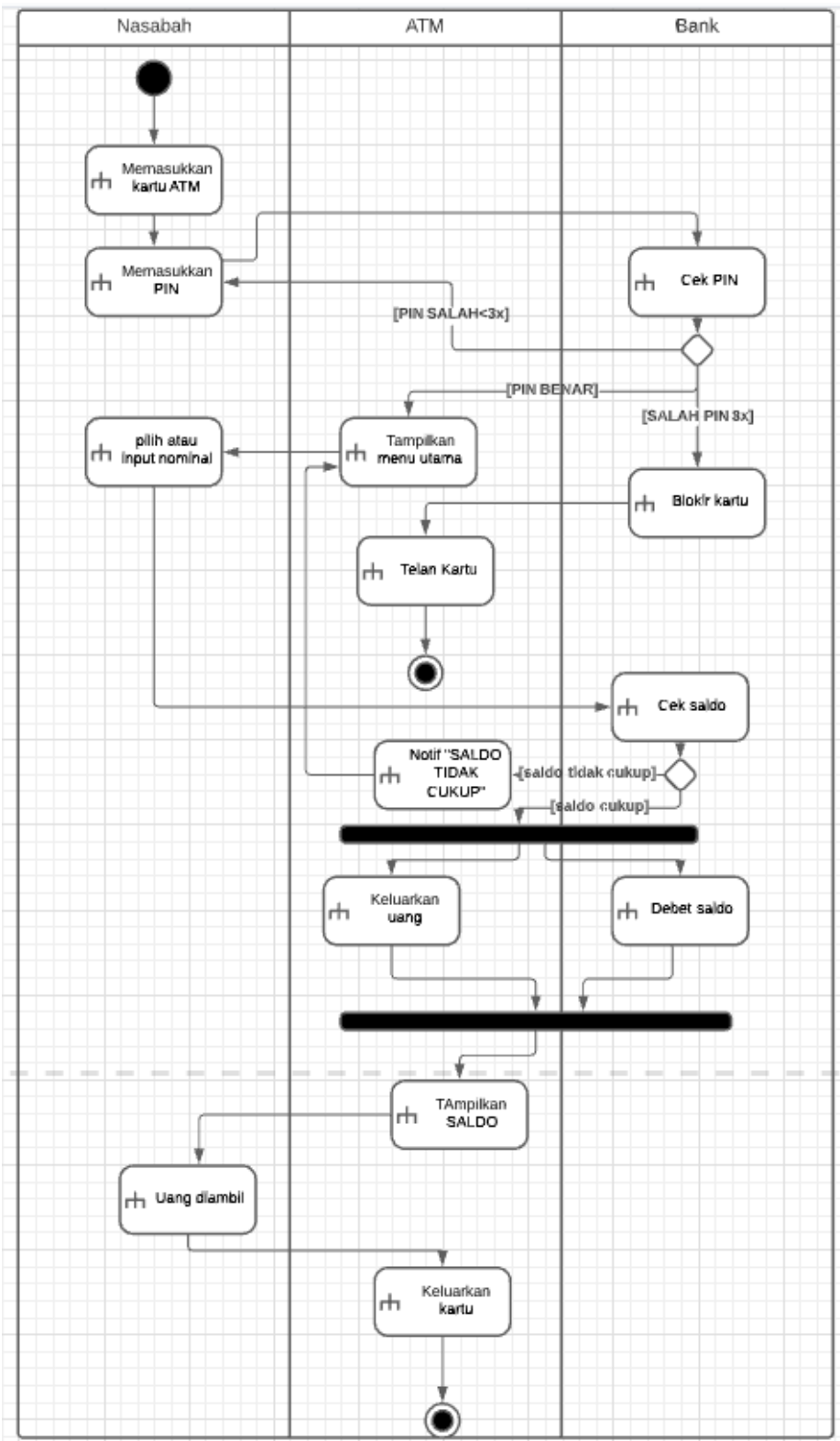
Alternate Flows	A1: Customer can enter duration instead of departure date, go to step 2.1 [A2] A2: Failed date check BR1. Notify error to Customer, go to step 2 A3: Complying with BR2, System determines that required rooms are not available, System upgrades one or more room types, go to step 2.1[A4] A4: No further upgrades available. Notify message to Customer, go to step 2 A5: Rooms offered are declined, go to step A9 A6: Customer already exists, Customer enters customer name and zip code, System searches for matching customers, notifies Customer of matching customers, Customer selects correct customer details, go to step 3.2 [A8]
Alternate Flows (continued)	A7: Payment guarantee fails. Notify message to Customer, go to step 3.2 A8: Existing customer not found, go to step 3.1 A9: Reservation not confirmed, reservation deleted, use case ends <i>At any time:</i> Customer may cancel the use case, use case ends [A9] <i>After use case inactivity of 10 minutes:</i> use case ends [A9]

Business Rules (BR)	<i>BR1:</i> The arrival date must not be before today's date, and the departure date must be after the arrival date <i>BR2:</i> Overbooking is not allowed <i>BR3:</i> Reservations with assigned rooms but no payment guarantee have a status of "held" <i>BR4:</i> The quoted price is the sum of the base price of the room types after applying BR5 and BR6 <i>BR5:</i> Seasonal Adjustment can be applied if reservation dates are applicable <i>BR6:</i> Offer adjustments can be applied if reservation qualifies <i>BR7:</i> Reservations with "held" status can be deleted <i>BR8:</i> Reservations with a status of "confirmed" must be linked to a payment guarantee and a customer <i>BR9:</i> Reservation must not exist without being linked to at least one room
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ACTIVITY DIAGRAM



Contoh Activity Diagram "Pengambilan Uang di ATM"



Contoh Kasus

Gojek:

<https://gelegakzaki.wordpress.com/2016/11/27/diagram-dan-analisa-baimana-go-jek-berjalan/>

 GO-RIDE	 GO-CAR	 GO-FOOD	
 GO-MART	 GO-BOX	 GO-MASSAGE	
 GO-GLAM	 GO-TIX	 GO-BUSWAY	
 GO-MED	 GO-AUTO	 GO-PULSA	

